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Interaction effects of night-time temperature and PM_{2.5} on preterm birth in Huai River Basin, China

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ABSTRACT

Nocturnal temperature is observed increasing with global warming. However, evidence on night-time non-optimal temperature on the risk of preterm birth (PTB) is limited, and the potential interactions with air pollution on PTB has not been well clarified. We therefore conducted a population-based retrospective cohort study to evaluate the effect of night-time temperature extremes on the risk of PTB and its interaction with air pollution. Records of 196,780 singleton births from 4 counties in Huai River Basin (2013–2018) were obtained. Gridded data on night-time temperature were collected from a high-quality Chinese Air Quality Reanalysis dataset. We used a multivariate logistic regression to evaluate the effects of night-time heat and cold exposure on the risk of PTB as well as its subtypes. Potential interactions between night-time temperature extremes and fine particulate matter < 2.5 µm (PM_{2.5}) were examined using the relative excess risk due to interaction (RERI). We found that the risk of PTB was positively associated with third trimester night-time extremely heat and cold exposure, with adjusted OR of 1.898 (95 %CI: 1.655–2.177) and 2.044 (95 %CI: 1.786–2.339). Similar effects were observed for PTB subtypes, moderately PTB (mPTB) and very PTB (vPTB). Synergistic effects (RERI greater than 0) of each trimester night-time temperature extremes exposure and PM_{2.5} on PTB were observed. We identified consistent positive interactions between night-time temperature extremes and PM_{2.5} on mPTB. No significant interaction of night-time temperature extremes and PM_{2.5} on vPTB was found. In conclusion, this large retrospective cohort study found that third trimester night-time heat and cold exposure significantly increased the risk of PTB and its subtypes. There is a synergistic effect between night-time temperature extremes and high PM_{2.5} levels on PTB and mPTB. In the context of climate warming, our results add new evidence to the current understanding of night-time non-optimal temperature exposure on PTB.

1. Introduction

Preterm birth (PTB) is one of the leading causes of death for children under 5 (Liu et al., 2016). According to World Health Organization (WHO), PTBs can be classified as moderate PTB (mPTB, 32–37 weeks), very PTB (vPTB, 28–32 weeks) (WHO 2018). PTB babies are at a higher risk of physiological and neurodevelopmental disorders in later life (Dance 2020; Sjöbom et al., 2021), such as alteration of bone structures, lipidome and apolipoprotein profiles, reduced docosahexaenoic acid availability for brain development and brain damage (Küster et al.,

2022; Perrone et al., 2022; Sjöbom et al., 2021), causing a substantial burden on families and healthcare systems (Walani 2020). It was estimated that the global PTBs were appropriately 15 million, and the rates are increasing in many regions, which mainly located in Asia and sub-Saharan Africa (Chawanpaiboon et al., 2019; Walani 2020; WHO 2012). The etiology of PTB has not been well clarified. Given the detrimental influence and high incidence of PTB, understanding the potentially modifiable triggers of PTB remains a priority for clinicians and public health professionals.

Climate warming has been an unequivocal fact in both observations

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and model simulations (IPCC 2022; Xu et al., 2018). In the context of climate warming, nocturnal temperature is increasing with global warming (Donat et al., 2013), the elevated night-time temperature is of specific concern because of lacking relief from high temperatures at night (Donat et al., 2012; Kovats et al., 2008; Laaidi et al., 2012; Murage et al., 2017; Royé 2017; Tollefson 2021). Thermal stress from night-time temperature is aggravated by the fact that the human body is prevented from nocturnal rest. Epidemiological studies have noted the particularly dangerous nature of night-time temperature extremes exposure (Murage et al., 2017; Royé 2017; Zhong et al., 2018).

Associations between temperature and health outcomes are typically non-linear (Gasparrini et al., 2015; Weinberger et al., 2017). There is no available consistent non-optimal temperature definition, so present studies mostly used different percentile to identify high and low non-optimal temperatures (Jiang et al., 2021; Li et al., 2018; Ren et al., 2022), which has been added as a risk factor of public health (Lancet 2020). Previous studies suggest that temperature extremes may increase the risk of PTB through increasing placental oxidative stress and systemic inflammation (Joo et al., 2021; Karahoda et al., 2021) or the sleep reduction during pregnancy (Acharya et al., 2022; Altena et al., 2022). Pregnant mothers are particularly vulnerable to non-optimal temperature and at a high risk of adverse perinatal outcomes, such as PTB (Balbus et al., 2009; Chersich et al., 2020; USAID 2019). Previous study observed that pre-labour spontaneous rupture of membranes, a major cause of PTB, generally occurred at night (Ngwenya et al., 2004), indicating that night-time temperature extremes may be associated with PTB. However, evidence is limited. The effect of night-time temperature on health may mainly impact populations less likely to heat the home or use air-conditioner (Sera et al., 2020). Therefore, we chose to perform our research in rural area, where the air-conditioning utilization rate is far below the national level (Gong et al., 2021).

In parallel, air pollution is becoming a health threat globally (Lancet 2020), perinatal exposure to PM_{2.5} is increasingly recognized as a hazard for PTB through similar mechanisms with temperature extreme exposure such as placental inflammation and oxidative stress (Bekkar et al., 2020; Feng et al., 2016; Kirwa et al., 2021; Yu et al., 2022). Previous studies have demonstrated synergistic effects of temperature extremes and PM_{2.5} on PTB (Kwag et al., 2021a; Kwag et al., 2021b; Wang et al., 2020a). However, whether nocturnal temperature extremes and PM_{2.5} could synergistically heighten the PTB risk is still unclear. The health impact associated with joint exposure to night-time extreme temperatures and air pollution may be larger than the risk estimated each alone. Air pollution and climate change share the similar source. Due to the high outdoor air pollution levels and the increasing night-time temperature, assessing the joint health impact should be performed. Furthermore, investigating the potential combined effects may contribute to better understanding how to minimize the potential adverse effect of the increasing exposure to night-time temperature extremes and PM_{2.5} on PTB, and a joint intervention during pregnancy may lead to more important benefits than isolated efforts (Kloog 2019; Sun et al., 2020).

We hypothesized that perinatal exposure to night-time temperature extremes is associated with PTB, and that it may exert a synergetic effect with PM_{2.5} on the association. In this study, we aimed to test this hypothesis using a large retrospective cohort study with spatially detailed data on temperature and air pollutants.

2. Materials and methods

2.1. Study population

A population-based retrospective cohort study was performed using data collected from Births Monitoring System from 2013 to 2018. Four counties in south, north, east, and west regions of Huai River Basin, China were randomly selected from each region, including: Shenqiu, Xiping, Luoshan and Fugou counties. All residents living in the four

counties were included. We performed the following exclusion criteria: First, we excluded births with <20 or more than 44 gestational weeks (Sun et al., 2019b). Second, we included singleton births, therefore, multiple pregnancies were not eligible (Kwag et al., 2021a). Third, we also excluded participants with age <13 or more than 50 years old (Shao et al., 2014; Song et al., 2011; Wang et al., 2020a) (Figure S1).

2.2. Outcome

Gestational age at birth was defined by combining clinical ultrasound examination and last menstrual period. Births at < 37 gestational weeks were defined as PTBs. We further categorized PTBs into moderate PTB (mPTB, 32–37 weeks) and very PTB (vPTB, 28–32 weeks) according to WHO recommendations (WHO 2018). Information on maternal address and age, date of birth, gestational age at delivery, sex of fetus, mode of delivery, and parity from all included mothers were also extracted from Births Monitoring System.

2.3. Exposure assessment

In the process of exposure assessment, we firstly obtained records on ambient temperature from the Chinese Air Quality Reanalysis datasets (CAQRA), a high-quality dataset that gridded at a 15 km × 15 km spatial resolution and at 1-hour temporal resolution from more than a thousand monitoring sites with a good performance in magnitude and variability. The modeling method and validation have been described previously (Kong et al., 2020). Second, each pregnant woman's residential address during pregnancy was collected and transformed into longitude and latitude by Baidu Map pickup coordinate system tool (Jiang et al., 2021). Last, each pregnant woman's residential address was then matched into the CAQRA model according to the longitude and latitude (Chen et al., 2021; Zhang et al., 2022a). Night-time for each day were considered as period from 20:00–07:00 in this study (Zhong et al., 2018). We derived the exposure of nocturnal temperature at each gestational week by averaging daily night-time temperature in every 7 days. The 1st day and the next 2nd – 7th days means a gestational week (Wang et al., 2018). Then we estimated night-time temperature in each trimester based on the weekly level: 1st (1–12 weeks), 2nd (13–27 weeks) and 3rd trimester (28 weeks until birth). Data on PM_{2.5} (μg/m³) and relative humidity (%) were simultaneously collected from the Chinese Air Quality Reanalysis datasets with same spatial and temporal resolution. We estimated PM_{2.5} and relative humidity exposure using same method for participants in each trimester like we did for night-time temperature.

There is no universally consistent temperature extremes definition (Dalugoda et al., 2022; Xu et al., 2016). Present studies have usually used a series of percentiles of temperature to define the high and low temperature exposure. We converted averaged night-time temperature to percentiles based on the distribution (Basagaña et al., 2021; Sun et al., 2019a). In this study, a series of percentiles were used as thresholds to determine temperature extremes (Dalugoda et al., 2022). The 95th, 90th and 75th percentiles of nocturnal temperature were defined as the extremely, moderately, and less heat exposure, while 5th, 10th, and 25th percentiles as extremely, moderately, and less cold exposure (Jiang et al., 2021; Li et al., 2018; Ren et al., 2022).

2.4. Statistical analysis

Analysis in this study was divided into two stages. First, we treated PTB and its subtypes as dichotomous outcomes, therefore, logistic regression models were used to investigate independent effects of exposure to night-time temperature extremes and PM_{2.5} on PTB (Gong et al., 2021; Zhong et al., 2018). We used odds ratio (OR) and 95 % confidence intervals (CIs) at 5th, 10th, 25th, 75th, 90th and 95th percentiles (defined as extremely, moderately, less cold and less, moderately, extremely heat, respectively) of night-time temperature in each trimester of pregnancy. ORs and 95 % CIs for a 10 μg/m³ PM_{2.5}

increment on the risk of PTB were calculated.

In the second stage, in order to explore whether combined effects of temperature extremes and PM_{2.5} on PTB were more or less than additive, we used relative excess risk due to interaction (RERI) (Lee et al., 2019), RERI greater than 0 means a positive interaction between temperature extremes and PM_{2.5} on the additive scale, which implying synergism between the two exposures than additive, while RERI < 0 indicates less than additive combined effects. To present the modifying effects of PM_{2.5} by temperature extremes more intuitively, we calculated strata-specific ORs and 95 % CI in each trimester (Jiang et al., 2021; Kwag et al., 2021a). Similar risk models were built for mPTB and vPTB as we did for the overall PTBs.

Potential confounding factors included maternal age (considered as a continuous variable), parity (classified as 0, 1 and ≥ 2), infant sex (male/ female), delivery pattern (caesarean or vaginal), season of conception (spring, summer, autumn, and winter), and socioeconomic status (SES). We did not obtain the individual-level SES information of each participant, like family income, education, and health conditions, which are important variables. Hence, we collected information on gross domestic product, doctors, hospital beds and schools in each county and divided into high, moderate, and low levels (Jain 2010; Yu et al., 2020; Zhang et al., 2022b). Relative humidity and daytime temperature exposure were also controlled for.

Sensitivity analyses were conducted. First, to examine the impact of adjusting for confounding, we repeated the analysis to investigate the associations without adjusting for mater's age, parity, infant sex, SES, mode of delivery, season of conception, relative humidity, daytime temperature. Second, we additionally adjusted for SO₂, NO₂, CO and O₃ in the model. Third, we also performed sensitivity analysis by excluding stillbirths that might have complicated causes and possibly bias our findings (Jiang et al., 2021). Forth, considering the geographical variation, we further conducted analysis for each county to examine the effects of night-time temperature on PTB, and then used meta-analysis to test the robustness of the pooled estimates (Ren et al., 2022; Sun et al., 2019b). We performed all analysis with R (Version number 3.4.0, R Development Core Team 2018), packages "glm" in R software were used. Significance level was set at $P < 0.05$.

2.5. Ethical approval

This is a retrospective cohort study, no personal contact with the participants. Therefore, informed consent was not collected. Additionally, the medical ethics committee of Zhengzhou University granted approval for this study.

3. Results

Our analysis included 196,780 singleton births, of which 2.2 % were PTBs (Table 1). Characteristics of PTBs and non-PTBs are shown in Table 1. The mean gestational age (SD) at delivery was 34.5 (2.3) weeks for PTBs and 39.4 (1.0) weeks for term births. The maternal age (mean and SD) for all participants was 26.2 ± 4.7 years old. Older (27.2 ± 5.5 years), higher parity (parity ≥ 2) mothers had a significant higher risk of PTB. Participants with a male baby had a higher PTB risk. PTB was more common among those resided in a lower SES. The mode of delivery among PTBs was more likely to be vaginal. The percentage of conception season was not significantly differed between the two groups.

Table 2 indicates distributions of night-time temperature and PM_{2.5} in all three trimester of pregnancy among participants. The median (50th percentile) and range (25th to 75th percentile) was 17.1 (8.2–24.9), 17.3 (8.8–24.2) and 16.1 (6.9–23.9) °C for night-time temperature in first, second and third trimester (Table 2). The mean concentration of PM_{2.5} exposure in each trimester were 94.5 $\mu\text{g}/\text{m}^3$, 99.0 $\mu\text{g}/\text{m}^3$, and 90.2 $\mu\text{g}/\text{m}^3$. The distributions of daytime temperature and other air pollutants were described in Table S1.

The adjusted OR and 95 %CI of night-time heat and cold temperature

Table 1

Characteristics of participants in present study.

Characteristics	Preterm birth ^a (n = 4,257)	Term birth (n = 192,523)	Total birth (n = 196,780)	P ^b
Gestational age (weeks (mean $\pm$ SD))	34.5 (2.3)	39.4 (1.0)	39.3 (1.3)	< 0.001
Maternal age (years, mean (SD))	27.2 (5.5)	26.2 (4.6)	26.2 (4.7)	< 0.001
Parity, n (%)				
0	435 (10.2)	27,797 (14.4)	28,232 (14.3)	< 0.001
1	1,929 (45.3)	87,663 (45.5)	89,592 (45.5)	
≥ 2	1,893 (44.5)	77,063 (40.0)	78,956 (40.1)	
Infant sex, n (%)				
Male	2,392 (56.2)	100,692 (52.3)	103,084 (52.4)	< 0.001
Female	1,846 (43.4)	91,824 (47.7)	93,670 (47.6)	
Missing	19 (0.4)	7 (0.004)	26 (0.02)	
SES ^c , n (%)				
Low	1,312 (30.8)	42,771 (22.2)	44,083 (22.4)	< 0.001
Medium	1,635 (38.4)	72,170 (37.5)	73,805 (37.5)	
High	1,292 (30.4)	76,923 (40)	78,215 (39.7)	
Missing	18 (0.4)	659 (0.3)	677 (0.3)	
Mode of delivery				
Caesarean	2,305 (54.1)	107,299 (55.7)	109,604 (55.7)	0.003
Vaginal	1,915 (45.0)	84,918 (44.1)	86,833 (44.1)	
Missing	37 (0.9)	306 (0.2)	343 (0.2)	
Season of conception, n (%)				0.11
Spring (March-May)	1,125 (26.4)	48,689 (25.3)	49,814 (25.3)	
Summer (June-August)	1,043 (24.5)	46,005 (23.9)	47,048 (23.9)	
Fall (September-November)	969 (22.8)	44,556 (23.1)	45,525 (23.1)	
Winter (December-February)	1,120 (26.3)	53,273 (27.7)	54,393 (27.6)	

Note: SES, socioeconomic status; SD, standard deviation; ^a Preterm birth indicates a live birth < 37 completed gestational weeks; ^b P value of chi-square test for categorical variable; ^c socioeconomic status at the county-level.

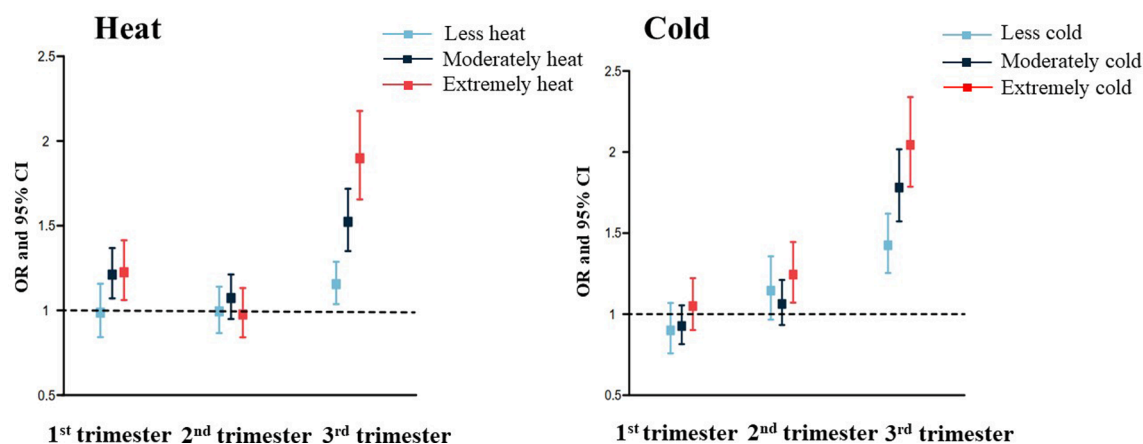
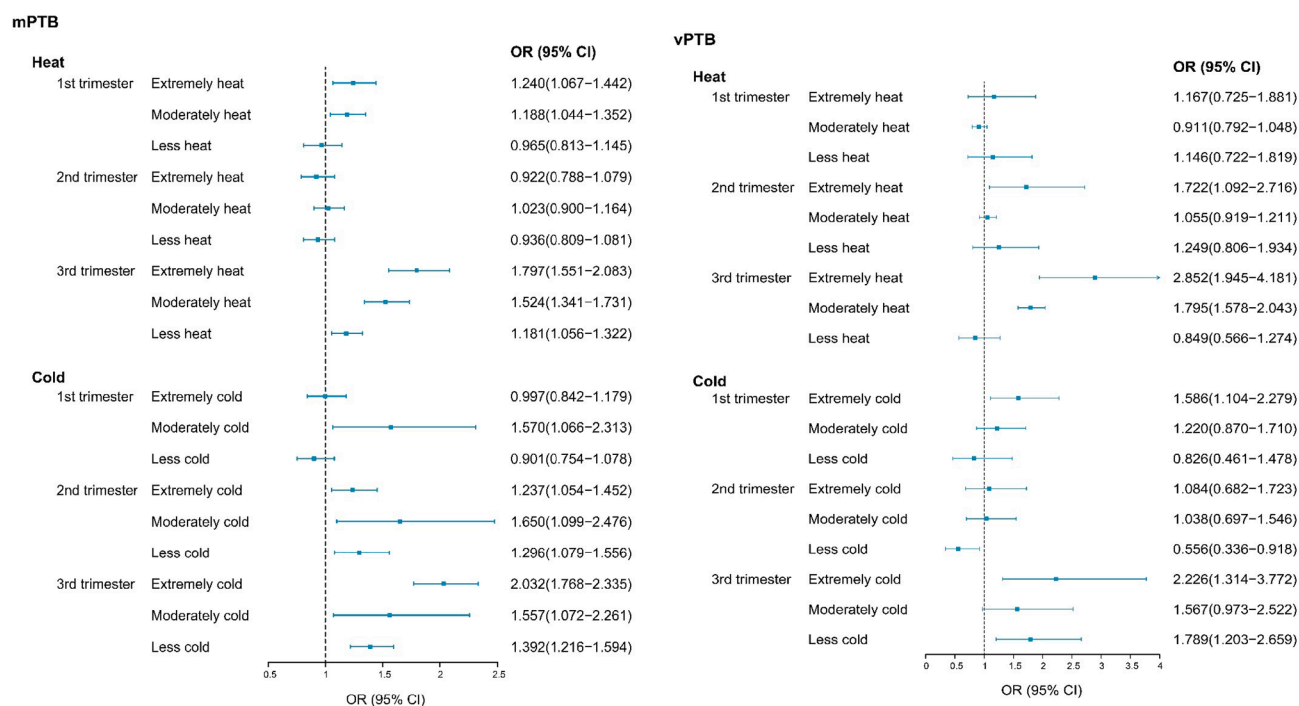
exposure on the risk of PTB in each trimester are presented in Fig. 1. Night-time heat and cold exposure were associated with PTB, positive associations were mainly in third trimester. As shown in Table S2, associations of extremely heat exposure with PTB were strongest in first and third trimester, the adjusted ORs and 95 % CIs were 1.225 (1.061–1.414) and 1.898 (1.655–2.177). We found that the effect of night-time cold exposure was mainly in third trimester, with strongest effect for extreme cold exposure (OR = 2.044, 95 %CI: 1.786–2.339). No substantial changes were found in sensitivity analysis (Table S3).

When stratified by gestational age, similar effects were observed for PTB subtypes, mPTB and vPTB, the effects were primarily in the third trimester and the strongest effect were observed for extremely heat and cold exposure (Fig. 2). Positive associations for PTB with a 10 $\mu\text{g}/\text{m}^3$ increase in second trimester PM_{2.5} were observed, with adjusted OR and 95 %CI of 1.014 (1.002–1.026). Effect from adjusted models were consistent with the crude model (Table S4). No significant associations between PM_{2.5} and mPTB and vPTB were observed, but the strongest effects were observed in second trimester of pregnancy (Table S5). We also explored the unhealthy level of PM_{2.5} on PTB, no significant results were observed (Figure S2).

The adjusted OR and 95 % CIs for the risk of PTB combined exposure

Table 2Percentile distribution of night-time temperature and PM_{2.5} by exposure period.

Exposures	Trimesters	Mean	SD	Min	5th	10th	25th	50th	75th	90th	95th	Max
Temperature (°C)	1st trimester	16.4	8.6	-1.5	3.5	4.4	8.2	17.1	24.9	27.2	28.0	29.3
	2nd trimester	16.5	8.0	-3.3	4.3	5.1	8.8	17.3	24.2	26.5	27.3	28.7
	3rd trimester	15.6	8.7	-2.4	3.1	3.9	6.9	16.1	23.9	26.8	27.7	33.6
PM _{2.5} (µg/m ³)	1st trimester	94.5	25.4	33.7	61.9	66.8	75.5	87.3	113.0	134.4	142.8	180.7
	2nd trimester	99.0	24.6	56.0	66.1	69.5	80.0	95.1	114.0	133.2	143.1	193.5
	3rd trimester	90.2	34.2	24.6	45.4	52.3	64.5	77.5	119.0	142.3	153.0	179.2

Note: PM_{2.5}, particles with aerodynamic diameters ≤ 2.5; SD: standard deviation.**Fig. 1.** Adjusted odds ratios of night-time temperature extremes exposure on PTB in three trimesters of pregnancy.**Fig. 2.** Adjusted odds ratios of moderate (mPTB) and very preterm birth (vPTB) associated with night-time temperature extremes in three trimesters of pregnancy.

to night-time temperature extremes and PM_{2.5} are presented in Table 3. Synergistic effect of night-time extremely heat exposure and PM_{2.5} in first, second and third trimester on PTB were observed, with RERI of 0.053 (95 %CI: 0.032–0.073), 0.041 (95 %CI: 0.018–0.064) and 0.063 (95 %CI: 0.011–0.115), similar effect was found for mPTB (Table 3).

PM_{2.5} acted synergistically with night-time extreme cold exposure in the first and second trimester of pregnancy, with RERIs from adjusted models of 0.051 (95 %CI: 0.033–0.068) and 0.046 (95 %CI: 0.031–0.062). We observed interaction effect of night-time extreme cold and PM_{2.5} exposure in all three trimester on mPTB. Results from the

Table 3Adjusted odds ratios and relative excess risk due to interaction of night-time temperature extremes and PM_{2.5} on PTB in three trimesters of pregnancy.

Temperature extremes	Exposure period		PM _{2.5} (μg/m ³)	PTB		mPTB		vPTB	
				OR	95 % CI	OR	95 % CI	OR	95 % CI
Extremely heat	1st trimester	No	< 87.3	1		1		1	
		No	≥87.3	0.991	0.931–1.055	0.991	0.927–1.060	0.992	0.820–1.201
		Yes	< 87.3	1.202	0.997–1.448	1.190	0.977–1.449	1.412	0.799–2.494
		Yes	≥87.3	1.241	1.015–1.518	1.290	1.048–1.589	0.857	0.393–1.870
		RERI		0.053	0.032–0.073	0.053	0.031–0.075	0.053	–0.007–0.112
	2nd trimester	No	< 95.1	1		1		1	
		No	≥95.1	0.966	0.907–1.029	0.969	0.906–1.035	0.969	0.797–1.179
		Yes	< 95.1	0.875	0.709–1.080	0.831	0.664–1.041	1.259	0.681–2.326
		Yes	≥ 95.1	1.302	1.106–1.532	1.154	0.965–1.380	2.841	1.910–4.224
		RERI		0.041	0.018–0.064	0.038	0.014–0.063	0.065	–0.002–0.133
	3rd trimester	No	< 77.5	1		1		1	
		No	≥ 77.5	0.978	0.918–1.042	0.995	0.930–1.064	0.858	0.703–1.048
		Yes	< 77.5	1.670	1.393–2.003	1.725	1.417–2.099	1.658	1.037–2.652
		Yes	≥ 77.5	1.936	1.627–2.303	1.971	1.631–2.381	2.117	1.367–3.278
		RERI		0.063	0.011–0.115	0.069	0.017–0.120	–0.104	–0.392–0.185
Extremely cold	1st trimester	No	< 87.3	1		1		1	
		No	≥87.3	0.955	0.897–1.017	0.954	0.892–1.019	0.960	0.789–1.169
		Yes	< 87.3	1.141	0.950–1.370	1.078	0.882–1.319	1.598	1.016–2.513
		Yes	≥87.3	1.112	0.917–1.347	1.051	0.851–1.299	1.731	1.098–2.728
		RERI		0.051	0.033–0.068	0.049	0.031–0.066	0.069	–0.046–0.183
	2nd trimester	No	< 95.1	1		1		1	
		No	≥95.1	0.985	0.925–1.049	0.979	0.916–1.046	1.062	0.877–1.285
		Yes	< 95.1	1.002	0.821–1.221	0.958	0.773–1.188	1.030	0.569–1.865
		Yes	≥ 95.1	1.155	0.943–1.415	1.170	0.944–1.449	0.840	0.410–1.720
		RERI		0.046	0.031–0.062	0.047	0.030–0.064	0.051	0.022–0.081
	3rd trimester	No	< 77.5	1		1		1	
		No	≥ 77.5	0.991	0.930–1.056	1.009	0.943–1.079	0.877	0.724–1.062
		Yes	< 77.5	1.910	1.604–2.274	2.045	1.707–2.450	0.889	0.426–1.854
		Yes	≥ 77.5	1.923	1.611–2.294	2.004	1.667–2.411	1.373	0.741–2.544
		RERI		0.068	–0.003–0.140	0.083	0.003–0.163	–0.002	–0.134–0.131

Note: PTB: preterm birth; mPTB: moderate PTB; vPTB: very PTB; RERI: relative excess risk due to interaction; Extreme heat was defined as 95th of night-time mean temperature, while 5th percentile of night-time mean temperature defined as the extreme low temperature exposure. The odds ratios were adjusted for maternal age, parity, infant sex, SES, mode of delivery, season of conception, relative humidity, daytime temperature.

interaction analysis for night-time moderate and less heat/cold exposure and PM_{2.5} are presented in Table S6 and S7. We found synergistic effects of moderately and less intense heat and PM_{2.5} in three trimesters on PTB and mPTB, similar effects were observed for moderately and less intense cold exposure.

4. Discussion

In this large population-based study, we investigated independent effect of night-time temperature extremes and its interaction with PM_{2.5} on PTB and its subtypes, mPTB and vPTB. Our results showed positive associations between third trimester night-time temperature extremes exposure and the risk of PTB and its subtypes. Interaction analysis suggested synergistic effects of night-time temperature extremes exposure and PM_{2.5} during pregnancy on PTB and mPTB.

The prevalence of PTB in this study was 2.2 %, which is consistent with previous studies in China (Ke et al., 2020; Zhang et al., 2015; Zhou et al., 2021). Under global warming, elevated night-time temperature is being paid close attention (Murage et al., 2017; Royé 2017). Numerous studies have explored effects of temperature extremes on PTB, however, evidence of night-time temperature extreme exposure on PTB is limited. Only one study found that heat exposure was associated with an elevated PTB risk, with higher risk for nocturnal temperature exposure (Zhong et al., 2018). No studies examined the susceptible windows of night-time temperature extremes and its interaction with air pollution is unclear. Wang et al. found that pregnant women at first and second trimester were susceptible to high temperatures while third trimester

were susceptible to low temperatures (Wang et al., 2020b). In a large population-based study from Brisbane, Australia, Li et al. (Li et al., 2018) defined the temperature extremes based on daily mean temperature and found that third trimester low and high temperature exposure increased the risk of PTB, He et al. also revealed that the association between extreme heat and PTB was stronger during late pregnancy (He et al., 2016). Consistently, we found that the susceptible windows of night-time extreme high and low temperature exposure and PTBs were primarily in the third trimester of pregnancy. Similar patterns were observed for PTB subtypes, mPTB and vPTB. In the context of climate warming, our findings might have important implications that the adverse effect of night-time temperature extremes should not be neglected in preventing PTBs.

It is plausible that temperature extremes can act interactively with PM_{2.5} to affect PTBs. In a retrospective study, Kwag et al. indicated a interaction between PM_{2.5} exposure and ambient temperature exposure, PM_{2.5} exposure in first trimester increased the risk of PTB among pregnant women exposed to low temperatures (Kwag et al., 2021b). In another study of seven major cities of Korea, Kwag et al. (Kwag et al., 2021a) found that the combined effect of heat wave and PM_{2.5} exposure during the 2nd trimester on PTB was greater than each exposure alone. In this study, we found synergistic effects of night-time heat/cold exposure and PM_{2.5} on PTB and mPTB in all three trimester. Our results indicated that pregnant women may should be aware of the risk associated with combined exposure to PM_{2.5} and night-time temperature extremes exposure.

The potential biological mechanism of night-time temperature

extremes on PTB risk is unclear. It was reported that heat may lead to inflammatory processes by increasing the production of proinflammatory cytokines during pregnancy, and heat stress could elevate the oxytocin and prostaglandin level, while low temperature is associated with peripheral vasoconstriction and hypertensive disorders during pregnancy, which may change uteroplacental perfusion and adversely impact fetus development (Carolan-Olah et al., 2014; Kloog 2019). In sheep models of heat stress exposure during pregnancy, significantly placental dysfunction was observed when pregnant ewes exposed chronically to high ambient temperatures in the laboratory from early to late gestation, which may cause growth restriction of fetuses (Limesand et al., 2018; Nair et al., 2021). Limesand et al. demonstrated that the majority of fetal growth restriction occurs during the final stages of gestation after placental growth restriction and decline in fetal glucose and oxygen concentrations over the final third of gestation were observed (Limesand et al., 2018), indicating third trimester are at high risk of PTB. Apart from unclear biological mechanism of night-time temperature extremes on PTB, other factors such as sleep loss may have potential modulating influence on the association. Exposure to heat stress during pregnancy may lead to sleep disruption (Altena et al., 2022). Acharya et al. found that pregnant mothers not having enough sleep had a higher risk of having a PTB baby born (Acharya et al., 2022). Whether sleep loss play a role in the effect of night-time temperature on PTB is unclarified, which warrants further research.

A 24 h rhythm of oxytocin and β -endorphin secretion in human pregnancy were investigated, Lindow et al. revealed that maternal blood levels of oxytocin had a 24 h rhythm with peak levels found at night-time (Lindow et al., 1996). Elevated oxytocin secretion could induce labor. The preterm labour onset has been also proved to have a 24 h rhythm with contractions starting mostly at night-time (Lindow et al., 2000). The above evidence indicated that night-time temperature extremes might promote the development of PTB. More investigations into the biological mechanism are warranted to elaborate our suggestive associations.

The main strength of this study is that we investigated the effect of night-time temperature extremes on the risk of PTB as well as its subtypes based on a large population-based retrospective cohort study, which adds new evidence of effects of night-time non-optimal temperature on the risk of PTB. The synergistic effect of $PM_{2.5}$ and night-time temperature extremes were explored in this study. Second, we conducted exposure based on gridded climate datasets, which could offer a meaningful method to exposure assessment of temperatures that may, in some areas, reduce the exposure measurement error (Spangler et al., 2019). Finally, we conducted sensitivity analyses to examine the robustness of our results, and consistent results were observed, which might increase the reliability of this study.

Several limitations should be considered when interpreting the findings. First, we used data from Births Monitoring System which are restricted to routinely registered information, maternal lifestyle factors such as smoking and alcohol consumption, gestational weight gain and pregnancy complications are not available due to the restriction of the Births Monitoring System, thus we were unable to fully adjusted for the confounding variables that might affect outcomes concurrently with variables of interest. Furthermore, factors like sleep loss related with night-time temperature extremes may modulate the influence of night-time temperature on PTB, however, such information is unavailable, leaving us unable to explore the potential modulating influence of sleep loss. Second, we collected maternal residential address, so we were not able to evaluate residential mobility for participants, measurement error may exist for not considering the activity pattern and residential mobility of pregnant mothers, however, the impact of this misclassification was likely toward the null. Previous study demonstrated that maternal residential mobility just have a minor impact (Warren et al., 2018), and Zhang et al. (Zhang et al., 2016) observed that only 3 % of pregnant women in China changed residential address. Third, maternal time spent indoors, outdoors, or time spent at workplace or home, air

purifier or the use of air conditioning or heating and other methods for mitigating temperature, were not obtained. These uncertainties about exposure settings should be considered when interpreting our findings. Forth, individual-level indicators of SES were not collected because of the unavailability of the data, instead, we adjusted for the area-level SES (Jain 2010; Yu et al., 2020). Last, this study only collected data from subtropical monsoon climate, our findings might be limited to other climate conditions. Consequently, more research is anticipated to be performed in multicounty large studies from other climate zones to corroborate the persistence and generalization of our results.

5. Conclusions

In this large retrospective cohort study, we found that third trimester night-time temperature extremes exposure increased the risk of PTB and its subtypes. Synergetic effects of night-time temperature extremes and $PM_{2.5}$ on the risk of PTB were observed, we found similar patterns for PTB subtype, mPTB. In the context of global warming, our findings add new evidence that night-time temperature extremes can independently and interact with $PM_{2.5}$ to increases the risk of PTB, which indicates that night-time temperature extremes exposure should not be neglected under global warming. It should be cautious when night-time temperature extremes exposure is accompanied by high $PM_{2.5}$ level. Further well-designed longitudinal studies incorporating maternal time-activity as well as collecting enough potential confounders in other climate zones and investigations into the possible molecular mechanisms are urgently warranted.

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CRedit authorship contribution statement

Huanhuan Zhang: Conceptualization, Data curation, Writing – original draft, Formal analysis. **Xiaoan Zhang:** Resources, Investigation. **Demin Feng:** Resources, Investigation. **Zhan Gao:** Data curation, Software, Methodology. **Yuting Gong:** Resources, Investigation. **Junxi Zhang:** Investigation. **Shuaixing Song:** Resources, Investigation. **Zengli Yu:** Conceptualization, Supervision, Resources, Investigation, Funding acquisition. **Cunrui Huang:** Writing – review & editing, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2023.107729>.

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